

AQSA SHEIKH

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PROFESSIONAL EXPERIENCE

Mindset Care Inc, Data Scientist

April 2024 – Present

- Created and led the full lifecycle development of an AI-powered intake and medical summarization service, integrating **OpenAI GPT-4o, LangChain, and Flask** to automate user interactions, eligibility assessments, and **medical data extraction** for legal and healthcare applications.
- Developed a scalable intake processing system leveraging **GPT-4o fine-tuned prompts, rule-based heuristics, and NLP pipelines** to extract structured user information, dynamically assess eligibility, and generate real-time decision insights.
- Engineered a **Slack-integrated AI assistant service** for **eligibility screening and decision-making**, connecting **Flask APIs, OpenAI LLMs, and DynamoDB** to provide context-aware responses, trigger intelligent follow-ups, and automate decision logic through a structured **JSON-based rule framework**.
- Developed an **AI-driven medical document processing service** using **Llama3-OpenBioLLM-8B, PyMuPDF, and AWS (S3, SageMaker, Databricks)** to extract key medical concepts, such as diseases, symptoms, medications, procedures, and anatomical structures, from unstructured medical data enabling legal brief generation for disability claims and large-scale medical data processing with NLP-based entity extraction.
- Built a **RAG-based AI agent** as a service using **LlamaIndex** to create **Chroma vectors** from medical files stored in **S3**, leveraging a fine-tuned **Llama3-OpenBioLLM-8B** model hosted on **AWS SageMaker** to ask structured medical questions and extract key insights about past medical history, symptoms, medications, hospital visits, treatment plans, and diagnoses.
- Developed an **API endpoint** enabling lawyers to **query** and retrieve relevant answers from large-scale medical data utilizing **Chroma vector** search for faster, context-aware responses. All medical data was stored in **DynamoDB** for efficient, real-time retrieval and analysis.
- Labeled and trained 47 custom classification models in **Azure Document Intelligence** to identify different types of disability documents stored in **Azure Blob Storage**, achieving **98% accuracy in classification**.
- Developed a custom extraction pipeline by labeling and training files to create **47 extraction models in Azure Document Intelligence**, enabling the automated extraction of relevant data based on document type with **96% accuracy in extraction**.
- **Automated** document ingestion and processing in **Databricks**, where files from **Azure Blob Storage** were classified using **Azure Document Intelligence**, extracted through **AI-driven models**, and stored in **DynamoDB** for real-time access as part of a scheduled **ETL job**.
- Created **real-time monitoring dashboards in Databricks** using **SQL queries** to generate charts that provided insights into document extraction performance, error rates, and ETL processing trends.
- **Developed real-time monitoring dashboards in Databricks**, equipping stakeholders with **interactive KPI widgets, daily vs. weekly success/failure trend analysis, and document status visualizations**. Leveraged **SQL queries** to track workflow efficiency, optimize model accuracy, and drive data-driven decision-making.

Compunnel Inc, Data/AI Engineer

June 2023- April 2024

- Establish and maintain resilient **ETL pipelines** for processing diverse pet health data, leveraging technologies like **AWS (Cloud Formation, S3, Lambda Glue, Redshift, Athena), Microservices, Databricks, SQL, Python and PySpark** for seamless ingestion, transformation, and loading.
- Created **Delta Tables** within **Databricks** and Implemented data ingestion into **Delta Tables**, followed by **Python-based transformations and SQL queries** for dashboard creation.
- Utilized **Delta Table's** advanced features, including ACID transactions, merge operation, schema evolution, and version control for robust data management.
- Developed a Contactform workflow using **AWS Step Functions, AWS SNS, AWS Lambda, and DynamoDB** to efficiently handle new user submissions.
- Created a pipeline leveraging **AWS Glue** components, by creating a **Crawler** to catalog pet health data, validating it through **Athena** queries for accuracy. Executed a meticulous **Glue Job** to convert CSV files to Parquet format, enhancing data efficiency and accessibility.
- Ensured table precision through **Athena** validation, seamlessly orchestrating the pipeline workflow. Integrated **Crawlers** to refresh tables, orchestrating **Glue Job** for format conversion.

- Integrated **PySpark** into the data processing pipeline of the pet healthcare application, harnessing its distributed computing capabilities for efficient large-scale data processing.
- Implemented optimization strategies using **CI/CD tools like Jenkins** for seamless integration, enhancing efficiency and ensuring error-free data operations.
- Trained **ML models (SVMs, Logistic Regression, Random Forests)** on pet sensor data to analyze sleep patterns, activity levels, and detect potential health disruptions. Developed a holistic health assessment model predicting risks and early signs of illness.
- Engineered **Flask-based API** functionalities using the **Transformers** library for dynamic health report generation and **Named Entity Recognition (NER)**, extracting medication names, symptoms, and dates from pet health records.
- Utilized **Streamlit** to create an intuitive interface featuring anomaly alerts and detailed health dashboards with actionable insights for pet's health status, using tools like **Tableau, pandas, Matplotlib, and Seaborn**.
- Developed a **Transformer-based question-answering system** enabling users to inquire about their pet's health.

Education Testing Services, Data Engineer/ AI Engineer

June 2022- March 2023

- **Contributed to engine and algorithm development** for English speaking, writing, and reading assessment models, resulting in improved accuracy and effectiveness of the models.
- Led the end-to-end **machine learning (ML) lifecycle** using **scikit-learn** for **Classification and Regression models**, encompassing problem framing, large-scale data collection, exploratory data analysis, model development, and performance measurement.
- Trained and deployed **ML models for prediction** generation and performance metric tracking, leveraging open-source tools including **Rater Scoring Modeling Tool (RSMTTool)** and **SKLL** for evaluating automated scoring models and generating customizable **HTML statistical reports**.
- Implemented and conducted automated unit tests with the **pytest and unittest** frameworks in **Python**, ensuring the accuracy and reliability validation of machine learning models.
- Developed **ETL** processes for ingesting data from diverse data sources into the data warehouse using **Python** and AWS Glue, resulting in improved data accessibility and analysis capabilities.
- Created **Airflow DAGs** to ensure **data integrity** and availability for training and scoring pipelines, while monitoring and **debugging** pipeline issues using web UI, logs, and metrics to ensure data quality and reliability.
- Utilized **SQL** and **PySpark** in **AWS Glue** to develop complex **data transformations**, enabling **data cleansing, normalization, and aggregation** from various sources like **XML, JSON and Restful API**
- Created interactive dashboards and reports in **Tableau** for business stakeholders, allowing for quick and effective decision making based on real-time data analysis.
- Designed and implemented **CI/CD** pipelines for data engineering and analytics projects using **Bamboo, Git, and Docker**, resulting in a **50%** reduction in deployment time and improved collaboration between teams.
- Independently designed and implemented data collection scripts in **Python and Bash** scripting. Applied advanced data cleaning techniques to aggregate and cleanse data, ensuring high-quality and consistent data for analysis.
- Utilized **JIRA** and **Confluence** contributing to **Sprint and user stories** creation and for documentation.

ClearOps Inc, Computer Programmer (NLP)

Oct 2019 – May 2022

Patent Issued | Patent: DAWN – Automated AI-Based Question Answering System | **Role-** Inventor

- **Researching, Studying, and building Deep Learning Model** in **Python** that will analyze data in Question Answering format using Deep Learning Framework like **BERT, Transformers, GPT-3, GPT-2, Roberta**
- Build multiple **Deep Learning Model** for NLP task such as Question classification, Reading Comprehension, knowledge Graph based Question Answering System, Semantic Analysis, Text Analytics
- Built Promise module using **NER** technology that extracted future promises done by companies while answering Security Questionnaire
- Build and integrated two **end-to-end Automated A.I powered Question Answering System** using Deep Learning frameworks like **BERT and Transformers** into the Web Application with **Feedback optimization**
- Extensively worked using machine learning libraries such as **Spacy, Stanza, NLTK, Seaborn, SciKit-Learn, SciPy** for machine learning
- Developed **NLP** models for **Named Entity Recognizer, Grammar & Spell Check, Entity Extraction, Sentiment Analysis, Question Generation, Text Summarization**
- Implemented an end-to-end data pipeline on **AWS Lambda** to extract, clean, preprocess CSV and Excel files for NLP model input, utilizing **Python** and loading processed data into **S3 buckets**.
- Worked on **data cleaning** and ensured data quality, consistency, integrity using **Pandas, NumPy, Python**

- Implemented and deployed an **NLP Automation Pipeline** on **AWS (S3, EC2, DynamoDB, Lambda)** and built **APIs** using **Flask** to serve various NLP models as endpoints for a web application.
- Developed and maintained automated data processing pipelines using **Jenkins**, allowing for continuous Integration and deployment (**CI/CD**) of **ETL** code changes. Created and managed **Jenkins jobs** for building, testing, and deploying data pipelines on **AWS** infrastructure.

PROJECTS

- **Build a Machine Learning Web App | Streamlit, Python, MLP:** [ML App Streamlit](#)
Build Machine Learning Web App using Python and Streamlit, enabling users to interactively explore and visualize data, create dashboards, and choose classification algorithms without coding required from users end.
- **Machine Learning Pipelines in PySpark | Python, Pyspark:** [ML PySpark](#)
Developed a PySpark pipeline with Random Forest, employed Cross Validation and Hyperparameter tuning for model parameter selection, and assessed performance through prediction creation in the pet wellness application.
- **Snowflake Market Share Snowsight Dashboard | Snowflake, SQL:** [Snowflake-Dashboard](#)
Developed a comprehensive Snowsight dashboard to analyze market share using SQL queries. Implemented efficient data management through virtual warehouses, databases, and tables in the Snowflake environment.
- **Daily Birth Forecasting Time Series | Python, NLTK, Scikit-learn, Flask:** [Disaster_message_classifier](#)
Built a multilabel classification model for disaster message categorization. Developed a web app with visualizations for message classification and implemented data visualization dashboards using Matplotlib and Plotly.
- **Files Manager | AWS (S3, Lambda, DynamoDB, API Gateway), Dash, Python:** [AWS_Visualization_Manager](#)
Pipeline for storing files along with meta-data, files can be searched and visualize using metadata

SKILLS

- **AI Frameworks and Libraries:** LangChain, LlamaIndex, OpenAI GPT-4o, GPT-3, T5, BERT, ChatGPT-4, Hugging Face Transformers, PyTorch, scikit-learn.
- **Programming and Scripting Languages:** Python, SQL, R, PySpark, Linux, Bash
- **Cloud & Data Engineering:** **AWS:** S3, SageMaker, Glue, Athena, Redshift
Azure: Blob Storage, Document Intelligence
- **ETL & Data Pipelines:** Databricks, Apache Airflow, AWS Glue, Azure Document Intelligence
- **Databases & Storage:** MySQL, PostgreSQL, Oracle, DynamoDB, ChromaDB
- **Machine Learning Frameworks:** PyTorch, Hugging Face Transformers, scikit-learn.
- **Data Visualization Tools:** Databricks Dashboard, Sigma, Tableau, Power BI, Seaborn, ggplot2
- **Data Warehousing:** Amazon Redshift
- **Version Control and CI/CD Tools:** Jenkins, GitHub, Bitbucket, Bamboo, Confluence, JIRA, Linear
- **Web & API Development:** FastAPI, Flask, Streamlit, Gradio

EDUCATION

Master's in information systems

New Jersey Institute of Technology, Newark, NJ, USA

Dec 2018

Bachelor's in biotechnology

Mumbai University, Mumbai, India

May 2016